Instrument Simulator Design Document

ERF Atmospheric Model

Ceilometer · Doppler Lidar · Microwave Radiometer

ERF Development Team

May 1, 2026

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1 Overview

This document describes the physics, algorithms, and software design of three ground-based remote sensing instrument simulators implemented in the ERF atmospheric model. The simulators provide forward operators that translate model state variables into synthetic instrument observables, enabling direct comparison with field campaign measurements.

All three simulators are implemented as header-only C++ structs in `Source/IO/ERF_InstrumentSimulators.H`. They are activated via `ParmParse` parameters in the input file and written on a user-specified timestep interval.

1.1 Instruments

Instrument	Make/Model	Wavelength	Primary Observables
Ceilometer	Vaisala CL-31	905 nm	$\beta_{\text{att}}(z,t)$, cloud base height
Doppler Lidar	Halo Photonics StreamLine XR+	1550 nm	$w(z,t)$, $\mathbf{U}(z,t)$, β_{att}
MWR	Radiometrics MP-3000A	22–58 GHz	$T(z,t)$, $q_v(z,t)$, LWP, PWV

1.2 Microphysics Compatibility

The backscatter forward model adapts automatically to the active microphysics scheme:

- **Bulk schemes (Kessler, Morrison, SAM, SatAdj, WSM6, Null)** — Use a fixed effective radius r_{eff} with bulk LWC/IWC extracted from `RhoQ2_comp` (liquid) and `RhoQ3_comp` (ice) in the conserved-state array (Section 2.2).
- **SBM** (Spectral Bin Microphysics) — Eulerian bin-resolving scheme; backscatter uses per-bin number concentration N_b [m^{-3}] and bin-centre radius r_b, identical to the `sbm_N_cloud` and `sbm_reflectivity` plotfile variables (Section 2.3).
- **SDM** (Super-Droplet Method) — Lagrangian particle-based scheme; backscatter uses each super-droplet's physical radius r_p and multiplicity ξ_p (Section 2.3).

The MWR simulator uses only bulk thermodynamic variables and behaves identically for all microphysics schemes.

2 Ceilometer Simulator (Vaisala CL-31, 905 nm)

2.1 Instrument Description

The Vaisala CL-31 is a single-lens backscatter lidar operating at $\lambda = 905\,\text{nm}$. It transmits short laser pulses and measures time-gated return signals. The primary observable is the **attenuated backscatter** coefficient $\beta_{\text{att}}(z,t)$ [$\text{m}^{-1}\,\text{sr}^{-1}$] as a function of height and time.

2.2 Simple Backscatter Forward Model (bulk schemes)

2.2.1 Volume Extinction Coefficient

For cloud droplets with effective radius $r_{\text{eff}} \gg \lambda$ (geometric optics regime, valid for $r_{\text{eff}} \gtrsim 5\,\mu\text{m}$):

$$\sigma_{\text{ext}}(z) = \frac{3}{2}\,\frac{\text{LWC}(z)}{\rho_w\, r_{\text{eff}}(z)} \quad [\text{m}^{-1}] \tag{1}$$

where:

$$\begin{aligned}
\text{LWC}(z) &= \rho_{\text{air}}(z)\, q_c(z) \quad [\text{kg m}^{-3}] \\
\rho_w &= 1000 \text{ kg m}^{-3} \quad \text{(liquid water density)} \\
r_{\text{eff}}(z) &= \frac{\int_0^\infty n(r)\, r^3\, dr}{\int_0^\infty n(r)\, r^2\, dr} \quad \text{(third/second moment of size distribution)}
\end{aligned}$$

For ice clouds the same form applies using IWC and $r_{\text{eff},i}$. In the absence of size-distribution information (all bulk schemes), the defaults are $r_{\text{eff}} = 10\,\mu$m (liquid) and $r_{\text{eff},i} = 30\,\mu$m (ice), both user-configurable.

Both liquid and ice contributions to extinction and backscatter are summed before applying the two-way optical depth:

$$\sigma_{\text{tot}}(z) = \sigma_{\text{liq}}(z) + \sigma_{\text{ice}}(z) \tag{2}$$

2.2.2 Volume Backscatter Coefficient

The lidar ratio S_L relates backscatter to extinction:

$$\beta(z) = \frac{\sigma_{\text{ext}}(z)}{S_L} \quad [\text{m}^{-1}\,\text{sr}^{-1}] \tag{3}$$

Adopted lidar ratio values:

Target	Wavelength	S_L
Liquid cloud	905 nm	18.8 sr
Ice cloud	905 nm	25.0 sr
Liquid cloud	1550 nm	20.0 sr
Ice cloud	1550 nm	28.0 sr

2.2.3 Two-Way Optical Depth and Transmission

The cumulative two-way optical depth from the surface to height z:

$$\tau(z) = 2\int_0^z \sigma_{\text{ext}}(z')\, dz' \tag{4}$$

Discretized (trapezoidal rule, k = vertical grid index):

$$\tau_k = \tau_{k-1} + \left(\sigma_{\text{tot},k} + \sigma_{\text{tot},k-1}\right)\Delta z_k \tag{5}$$

2.2.4 Attenuated Backscatter

$$\boxed{\beta_{\text{att}}(z) = \beta(z)\,\exp\!\left[-\tau(z)\right]} \quad [\text{m}^{-1}\,\text{sr}^{-1}] \tag{6}$$

This is the quantity directly observed by the ceilometer.

2.3 Size-Resolved Geometric-Optics Backscatter (SBM and SDM)

When SBM or SDM microphysics is active, the extinction coefficient is computed from the actual droplet/crystal size distribution using the geometric-optics limit ($Q_{\text{ext}} = 2$, valid for $r \gg \lambda$, i.e. cloud droplets $r \gtrsim 5\,\mu$m). The same variables and bin radii are used as in the ERF plotfile diagnostics (`sbm_N_cloud`, `sbm_reflectivity`), ensuring consistency between plotfile and instrument-simulator outputs.

2.3.1 SBM Path

For each liquid (or ice) bin b with bin-centre radius r_b [m] and number concentration N_b [m^{-3}]:

$$\sigma_{\text{ext},b}(z) = N_b(z) \cdot 2\pi r_b^2 \quad [\text{m}^{-1}] \tag{7}$$

where N_b is read from `cons[RhoQ1_comp + idx_nc_start + b]` (two-moment SBM, N_b in m^{-3}), and $r_b =$ `GetDiameterGridHostUM()`$[b]/2$. The column extinction is the sum over all 33 liquid (or ice) bins:

$$\sigma_{\text{ext,liq}}(z) = \sum_{b=0}^{32} \sigma_{\text{ext},b}^{\text{liq}}(z), \qquad \sigma_{\text{ext,ice}}(z) = \sum_{b=0}^{32} \sigma_{\text{ext},b}^{\text{ice}}(z) \tag{8}$$

A one-moment fallback derives N_b from the mass density bin if `idx_nc_start` < 0.

2.3.2 SDM Path

For each super-droplet p in a column $(i_{\text{loc}}, j_{\text{loc}})$, its contribution to vertical cell k is accumulated on the GPU using atomic addition:

$$\sigma_{\text{ext}}(k) \mathrel{+}= \frac{\xi_p \cdot 2\pi r_p^2}{V_{\text{cell}}} \quad [\text{m}^{-1}] \tag{9}$$

where ξ_p is the multiplicity (real droplet count represented), r_p [m] is the physical droplet radius, and $V_{\text{cell}} = \Delta x\,\Delta y\,\Delta z$ is the grid cell volume. The vertical index k is obtained from `p.idata(ERFParticlesIntIdxAoS::k)`. Accumulation uses `amrex::Gpu::Atomic::AddNoRet` on a `Gpu::DeviceVector`, followed by device-to-host copy and `MPI_Reduce` to the IO processor. SDM does not separately tag liquid vs. ice phase; the full σ_{ext} is stored in `sigma_liq`, with `sigma_ice` $= 0$.

2.3.3 Data Pathway

The pre-computed σ columns are packaged into an `InstrMicrophySizeData` struct and passed to `write()` as an optional pointer:

```
struct InstrMicrophySizeData {
    Vector<Vector<Real>> sigma_liq;  // [site][k]  m-1
    Vector<Vector<Real>> sigma_ice;  // [site][k]  m-1
    bool empty() const { return sigma_liq.empty(); }
};
// In ERF::Evolve() -- only on write steps (avoids overhead every step):
ceil_mpdata = fill_sbm(ilocs, jlocs);   // or fill_sdm(...)
m_ceilometer_sim->write(..., ceil_mpdata.get());
```

Backscatter is computed via `InstrumentSimUtil::compute_beta_att_from_sigma()`, which applies the optical-depth integral and lidar-ratio conversion (equations 4–6). When `mpdata == nullptr` (any bulk scheme), the simulator falls back to the simple fixed-r_{eff} model (Section 2.2).

2.4 Mie Scattering Forward Model (Planned — Not Yet Implemented)

Note: The full Mie backscatter series is not yet implemented. When `backscatter_model = mie` *is specified, the code currently falls back to the size-resolved geometric-optics model of Section 2.3 (SBM/SDM) or the bulk fixed-r_{eff} model (bulk schemes). The section below documents the planned Mie algorithm for future implementation.*

2.4.1 Mie Efficiency Factors

For each SBM bin n with bin-center radius r_n at wavelength λ, the size parameter is:

$$x_n = \frac{2\pi r_n}{\lambda} \tag{10}$$

The Mie extinction and backscatter efficiency factors $Q_{\text{ext}}(x_n, m)$ and $Q_{\text{back}}(x_n, m)$ are computed via the Bohren & Huffman (1983) Mie series [2]:

$$Q_{\text{ext}} = \frac{2}{x^2} \sum_{l=1}^{N} (2l+1)\,\text{Re}(a_l + b_l) \tag{11}$$

$$Q_{\text{back}} = \frac{1}{x^2} \left| \sum_{l=1}^{N} (2l+1)(-1)^l (a_l - b_l) \right|^2 \tag{12}$$

where a_l, b_l are the Mie coefficients (complex Riccati-Bessel functions), $m = n_r - i n_i$ is the complex refractive index of water/ice.

Refractive indices used:

- Liquid water at 905 nm: $m = 1.328 - 1.67 \times 10^{-8} i$
- Liquid water at 1550 nm: $m = 1.318 - 1.18 \times 10^{-4} i$
- Ice at 905 nm: $m = 1.304 - 3.65 \times 10^{-9} i$

These tables are precomputed at initialization using the SBM bin diameters from `ERF_BinGrid.H` (`m_diameter_grid_host`).

2.4.2 Column-Integrated Mie Backscatter

$$\beta_{\text{Mie}}(z) = \sum_n N_n(z)\,\pi r_n^2\,Q_{\text{back},n} \quad [\text{m}^{-1}\,\text{sr}^{-1}] \tag{13}$$

$$\sigma_{\text{Mie}}(z) = \sum_n N_n(z)\,\pi r_n^2\,Q_{\text{ext},n} \quad [\text{m}^{-1}] \tag{14}$$

where $N_n(z)$ [m^{-3}] is the number concentration in bin n at height z. The attenuated backscatter is then:

$$\beta_{\text{att}}(z) = \beta_{\text{Mie}}(z)\,\exp\left(-2\int_0^z \sigma_{\text{Mie}}(z')\,dz'\right) \tag{15}$$

2.4.3 SDM (Lagrangian) Adaptation

For SDM, at each grid cell (i, j, k) the super-droplet radii are histogrammed into a temporary array matching the SBM bin grid. The number concentration per bin is estimated as:

$$N_n = \frac{1}{V_{\text{cell}}} \sum_{p\,:\, r_p \in [r_n^-, r_n^+]} \xi_p \tag{16}$$

where ξ_p is the multiplicity (real droplet count) of super-droplet p, and V_{cell} is the grid cell volume.

2.5 Cloud Base Detection

Cloud base height is the lowest level where β_{att} exceeds a threshold:

$$z_{\text{cb}} = \min\{z : \beta_{\text{att}}(z) > \beta_{\text{thresh}}\} \tag{17}$$

Default: $\beta_{\text{thresh}} = 3 \times 10^{-4}\,\text{m}^{-1}\,\text{sr}^{-1}$. Output is -9999 if no cloud is detected.

3 Doppler Lidar Simulator (Halo StreamLine XR+, 1550 nm)

3.1 Instrument Description

The Halo Photonics StreamLine XR+ is a coherent pulsed Doppler lidar at $\lambda = 1550\,\text{nm}$. It operates in multiple scan modes:

- **Vertical stare**: beam pointing zenith, measures $w(z, t)$ and $\text{CNR}(z, t)$
- **VAD scan**: conical scan at fixed elevation angle, retrieves $u(z, t)$, $v(z, t)$
- **Attenuated backscatter**: passive measurement of $\beta_{\text{att}}(z, t)$

Each mode is controlled independently via input parameters.

3.2 Vertical Stare Mode

For a zenith-pointing beam, the radial (Doppler) velocity equals the vertical wind component:

$$V_r(z) = w(z) \quad [\text{m}\,\text{s}^{-1}] \tag{18}$$

The cell-centred vertical velocity is the average of the two enclosing face values: $w_k = \frac{1}{2}(w_{k-1/2} + w_{k+1/2})$.

3.2.1 CNR Model Selection (`cnr_model`)

The Carrier-to-Noise Ratio (CNR) is output alongside $w(z, t)$ in the vertical-stare file. Two models are available.

Normalized model (default, `cnr_model = normalized`) A dimensionless range-normalized proxy convenient for qualitative comparison:

$$\text{CNR}(r) = \frac{\beta_{\text{att}}(r)}{r^2} \tag{19}$$

This requires `do_backscatter = true` to supply β_{att}; otherwise CNR is output as zero.

Physical model (`cnr_model = physical`) Based on the coherent Doppler lidar performance equation of Targ et al. (1991) [1]. The CNR is related to the attenuated backscatter through the instrument function $F(r)$:

$$\boxed{\mathrm{CNR}(r) = F(r)\ \beta_{\mathrm{att}}(r)} \tag{20}$$

where

$$F(r) = \frac{E\,\eta\,\lambda\,\pi\,\omega^2}{2\,h\,B_w\,r^2} \cdot \underbrace{\left[1 + \left(1 - \frac{r}{R}\right)^2 \left(\frac{\pi\omega^2}{\lambda r}\right)^2\right]^{-1}}_{\text{Gaussian beam focus correction}} \tag{21}$$

The parameters are:

Symbol	Parameter name	Meaning	Default (Halo XR+)
E	`pulse_energy`	Pulse energy [J]	1.5×10^{-6} J
η	`optics_efficiency`	Optical + detector efficiency [-]	0.1
λ	—	Wavelength [m] (fixed)	1.55×10^{-6} m
ω	`beam_radius`	$1/e^2$ beam radius at aperture [m]	0.0225 m
h	—	Planck constant [J s] (fixed)	6.626×10^{-34} J s
B_w	`bandwidth`	Receiver noise bandwidth [Hz]	50×10^6 Hz
R	`focus_range`	Radius of curvature at aperture [m]	10^9 m (collimated)

The factor $\pi\omega^2/(\lambda r)$ is the Rayleigh-range-to-range ratio. The focus correction equals 1 when $r = R$ (beam focused at range R) and approaches 1 everywhere for a collimated beam ($R \to \infty$).

3.3 VAD (Velocity-Azimuth Display) Mode

3.3.1 Radial Velocity Simulation

For a beam at elevation angle ϕ from horizontal and azimuth ψ (clockwise from North), the beam unit vector in East-North-Up coordinates is:

$$\hat{d} = (\cos\phi\cos\psi,\ \cos\phi\sin\psi,\ \sin\phi) \tag{22}$$

The simulated radial velocity (projection of wind onto beam):

$$\boxed{V_r(\psi, z) = u(z)\cos\phi\cos\psi + v(z)\cos\phi\sin\psi + w(z)\sin\phi} \tag{23}$$

Sampled at $N_\psi = 36$ azimuths: $\psi_m = m \times 10°$, $m = 0, 1, \ldots, 35$. Cell-centred u and v are computed by averaging the two enclosing face values.

3.3.2 VAD Retrieval

The radial velocity field is modeled as a truncated Fourier series in azimuth:

$$V_r(\psi, z) \approx A_0(z) + A_1(z)\cos\psi + B_1(z)\sin\psi \tag{24}$$

Coefficients are determined by least-squares fit over the N_ψ azimuths. The horizontal wind components are retrieved as:

$$w(z) = A_0(z)/\sin\phi \tag{25}$$

$$u(z) = A_1(z)/\cos\phi \tag{26}$$

$$v(z) = B_1(z)/\cos\phi \tag{27}$$

Horizontal wind speed and meteorological wind direction:

$$|V_h|(z) = \sqrt{u(z)^2 + v(z)^2} \tag{28}$$

$$\mathrm{WD}(z) = \mathrm{atan2}(-u, -v) \times \frac{180}{\pi} + 180 \quad \text{[degrees, meteorological convention]} \tag{29}$$

3.4 Attenuated Backscatter at 1550 nm

Same forward model as the ceilometer (Equation 1–6) but with $\lambda = 1550\,\mathrm{nm}$ lidar ratios ($S_L = 20\,\mathrm{sr}$ for liquid, 28 sr for ice). Both liquid and ice cloud contributions are included:

$$\beta_{\mathrm{att}}(z) = \left[\beta_{\mathrm{liq}}(z) + \beta_{\mathrm{ice}}(z)\right] \exp(-\tau_{\mathrm{liq}}(z) - \tau_{\mathrm{ice}}(z)) \tag{30}$$

4 Microwave Radiometer Simulator (Radiometrics MP-3000A)

4.1 Instrument Description

The Radiometrics MP-3000A is a passive microwave radiometer with 35 channels:

- **K-band**: 22 channels from 22.235 to 40.0 GHz — water vapor absorption
- **V-band**: 13 channels from 51.25 to 58.0 GHz — oxygen absorption complex

The instrument retrieves temperature profiles, water vapor profiles, LWP, and PWV using neural network algorithms applied to observed brightness temperatures.

4.2 Direct Profile Output

4.2.1 Temperature Profile

Model potential temperature θ is converted to absolute temperature via the Exner function:

$$T(z) = \theta(z)\,\Pi(z) \tag{31}$$

$$\Pi(z) = \left(\frac{p(z)}{p_0}\right)^{R_d/c_p} \tag{32}$$

where:

$$\begin{aligned}
\theta(z) &= (\rho\theta)(z)/\rho(z) \quad \text{from state array} \\
p(z) &= p_0(z) \quad \text{(base-state pressure, 1D column)} \\
p_0 &= 10^5\ \mathrm{Pa} \quad \text{(reference pressure)} \\
R_d &= 287.04\ \mathrm{J\,kg^{-1}K^{-1}} \\
c_p &= 1004.5\ \mathrm{J\,kg^{-1}K^{-1}}
\end{aligned}$$

4.2.2 Water Vapor and Cloud Liquid Profiles

$$q_v(z) = \frac{(\rho q_v)(z)}{\rho(z)} = \frac{\texttt{state}(i, j, k, \text{ RhoQ1_comp})}{\texttt{state}(i, j, k, \text{ Rho_comp})} \quad [\mathrm{kg\,kg^{-1}}] \tag{33}$$

$$q_c(z) = \frac{(\rho q_c)(z)}{\rho(z)} = \frac{\texttt{state}(i, j, k, \text{ RhoQ2_comp})}{\texttt{state}(i, j, k, \text{ Rho_comp})} \quad [\mathrm{kg\,kg^{-1}}] \tag{34}$$

Both q_v and q_c are output in $\mathrm{g\,kg^{-1}}$ in the profile file.

4.2.3 Column Integrals

Liquid Water Path (LWP):

$$\mathrm{LWP} = \int_0^{z_{\mathrm{top}}} \rho(z)\, q_c(z)\, dz \approx \sum_k \rho_k\, q_{c,k}\, \Delta z_k \quad [\mathrm{kg\,m^{-2}}] \tag{35}$$

Output in $\mathrm{g\,m^{-2}}$ (multiply by 1000).

Precipitable Water Vapor (PWV):

$$\mathrm{PWV} = \frac{1}{\rho_w} \int_0^{z_{\mathrm{top}}} \rho(z)\, q_v(z)\, dz \approx \frac{1}{\rho_w} \sum_k \rho_k\, q_{v,k}\, \Delta z_k \quad [\mathrm{mm}] \tag{36}$$

where $\rho_w = 1000\ \mathrm{kg\,m^{-3}}$ and the result in mm equals the column integral in $\mathrm{kg\,m^{-2}}$ (numerically).

4.3 Brightness Temperature Forward Model (Optional)

4.3.1 Gaseous Absorption

The total volume absorption coefficient at frequency f [GHz] and height z is:

$$\alpha(f, z) = \alpha_{\mathrm{H_2O}}(f, T, p, \rho_v) + \alpha_{\mathrm{O_2}}(f, T, p) + \alpha_{\mathrm{liq}}(f, T, \mathrm{LWC}) \quad [\mathrm{Np\,m^{-1}}] \tag{37}$$

Water vapor absorption (K-band) Simplified Rosenkranz (1998) model [3] with dominant line at 22.235 GHz plus a continuum term. The Van Vleck-Weisskopf lineshape is:

$$F_i(f) = \frac{f}{f_0} \left[\frac{\gamma}{(f - f_0)^2 + \gamma^2} + \frac{\gamma}{(f + f_0)^2 + \gamma^2} \right] \tag{38}$$

Line width γ depends on pressure and water vapor density following Rosenkranz (1998) Table 1.

Oxygen absorption (V-band) Simplified Liebe et al. (1993) [4] oxygen complex. Thirty dominant lines from 50.474 to 66.302 GHz are summed with pressure-broadened Van Vleck-Weisskopf profiles and line strengths scaled as $\theta^3\, p$.

Liquid water absorption Rayleigh approximation valid for $f \ll 300$ GHz:

$$\alpha_{\mathrm{liq}}(f) = \frac{6\pi f}{c} \frac{3\,\varepsilon''}{(\varepsilon_r + 2)^2 + (\varepsilon'')^2} \frac{\mathrm{LWC}}{\rho_w} \tag{39}$$

The complex permittivity $\varepsilon = \varepsilon_r - i\varepsilon''$ of liquid water uses the Ellison (2007) double-Debye parameterization [5].

4.3.2 Radiative Transfer

For zenith-pointing geometry, the upwelling brightness temperature is:

$$\boxed{T_B(f) = T_{\mathrm{bg}}\, e^{-\tau_{\mathrm{tot}}(f)} + \int_0^{\infty} T(z)\, \alpha(f, z)\, e^{-\tau(f, 0, z)}\, dz} \tag{40}$$

where:

$$T_{\text{bg}} = 2.75 \text{ K} \quad \text{(cosmic microwave background)}$$
$$\tau(f, z_1, z_2) = \int_{z_1}^{z_2} \alpha(f, z')\, dz'$$
$$\tau_{\text{tot}}(f) = \tau(f, 0, \infty)$$

Discrete form (summed over K vertical levels, τ_k = optical depth from surface to z_k):

$$T_B(f) \approx T_{\text{bg}}\, e^{-\tau_K} + \sum_{k=1}^{K} T_k \left(e^{-\tau_{k-1}} - e^{-\tau_k}\right) \tag{41}$$

4.3.3 MP-3000A Channel Frequencies

K-band (22 channels)		**V-band (13 channels)**	
Ch	f [GHz]	Ch	f [GHz]
1	22.235	23	51.250
2	22.500	24	51.760
3	23.035	25	52.280
4	23.835	26	52.800
5	25.000	27	53.340
6	26.235	28	53.860
7	28.000	29	54.400
8	30.000	30	54.940
9	31.400	31	55.500
10	32.000	32	56.020
11	32.400	33	56.660
12	33.000	34	57.290
13	34.000	35	58.000
14	35.000		
15	36.000		
16	36.500		
17	37.000		
18	37.500		
19	38.000		
20	38.600		
21	39.500		
22	40.000		

5 Input File Parameters

All parameters use the `erf.*` ParmParse prefix. Multiple instrument sites can be specified by providing equal-length lists for `i_loc` and `j_loc`; indices are 0-based cell indices.

Listing 1: Ceilometer input parameters (defaults shown)

```
# Ceilometer (Vaisala CL-31, 905 nm)
erf.ceilometer.do_sim              = true
```

```
erf.ceilometer.i_loc              = 32 48    # REQUIRED; 0-based i indices
    (list OK)
erf.ceilometer.j_loc              = 32 48    # REQUIRED; must match i_loc
   count
erf.ceilometer.output_interval    = 10       # write every N timesteps
erf.ceilometer.output_file        = ceil_    # prefix; ceil_00.txt,
   ceil_01.txt, ...
erf.ceilometer.backscatter_model  = simple   # "simple" (geometric optics)
    or "mie"
erf.ceilometer.r_eff_default      = 1.0e-5  # m; liquid effective radius
   fallback
erf.ceilometer.r_eff_ice_default  = 3.0e-5  # m; ice effective radius
   fallback
erf.ceilometer.beta_thresh        = 3.0e-4  # m-1 sr-1; cloud base
   detection threshold
```

Listing 2: Doppler Lidar input parameters (defaults shown)

```
# Doppler Lidar (Halo StreamLine XR+, 1550 nm)
erf.doppler_lidar.do_sim               = true
erf.doppler_lidar.i_loc                = 32      # REQUIRED; one or more
   sites
erf.doppler_lidar.j_loc                = 32      # REQUIRED
erf.doppler_lidar.output_interval      = 10
erf.doppler_lidar.output_file          = dl_
erf.doppler_lidar.backscatter_model    = simple  # "simple" or "mie"
erf.doppler_lidar.r_eff_default        = 1.0e-5  # m; liquid effective
   radius fallback
erf.doppler_lidar.do_vertical_stare    = true    # output w(z,t) and CNR  [
   default: true]
erf.doppler_lidar.do_vad               = false   # output u,v,wind speed/
   dir [default: false]
erf.doppler_lidar.vad_elevation_angle = 75.0    # degrees from horizontal
erf.doppler_lidar.do_backscatter       = false   # output beta_att(z,t)  [
   default: false]
# CNR model for vertical stare output:
erf.doppler_lidar.cnr_model            = normalized  # "normalized" (
   default) or "physical"
# Physical CNR parameters (Targ 1991) -- only used when cnr_model =
   physical:
erf.doppler_lidar.pulse_energy         = 1.5e-6  # E   [J]
erf.doppler_lidar.optics_efficiency    = 0.1     # eta [-]
erf.doppler_lidar.beam_radius          = 0.0225  # omega [m]  1/e^2 beam
   radius
erf.doppler_lidar.bandwidth            = 50.0e6  # Bw  [Hz]  receiver
   bandwidth
erf.doppler_lidar.focus_range          = 1.0e9   # R   [m]   (1e9 =
   collimated)
```

Listing 3: MWR input parameters (defaults shown)

```
# MWR (Radiometrics MP-3000A, 35 channels)
erf.mwr.do_sim             = true
erf.mwr.i_loc              = 32      # REQUIRED
```

```
erf.mwr.j_loc               = 32      # REQUIRED
erf.mwr.output_interval     = 10
erf.mwr.output_file         = mwr_
erf.mwr.do_brightness_temp = false   # true = also write simulated TB (35
    channels)
```

6 Output File Formats

All output files are ASCII text with comment headers (# prefix lines). Time rows are appended at each write interval.

6.1 Ceilometer: ceil_XX.txt

```
# ERF Ceilometer Simulator (Vaisala CL-31, 905 nm)
# i_loc=32  j_loc=32
# backscatter_model=simple
# Col 1: time [s]
# Col 2: height [m]
# Col 3: att_backscatter [m-1 sr-1]
# Col 4: cloud_base_height [m]  (-9999 = no cloud)
  1.000000e+01  2.500000e+01  1.234000e-06  2.500000e+02
  1.000000e+01  7.500000e+01  3.456000e-04  2.500000e+02
```

6.2 Doppler Lidar Stare: dl_XX_stare.txt

```
# ERF Doppler Lidar Simulator (Halo XR+, 1550 nm) -- vertical stare
# i_loc=32  j_loc=32
# cnr_model=normalized  (CNR = beta_att / r^2)
# Col 1: time [s],  Col 2: height [m]
# Col 3: w [m s-1],  Col 4: CNR [-]
  1.000000e+01  2.500000e+01  1.230000e-01  4.560000e-03
```

6.3 Doppler Lidar VAD: dl_XX_vad.txt

```
# ERF Doppler Lidar Simulator -- VAD mode (elev=75.0 deg, 36 azimuths)
# Col 1: time [s],  Col 2: height [m]
# Col 3: u [m s-1],  Col 4: v [m s-1]
# Col 5: wind_speed [m s-1],  Col 6: wind_dir [deg, met convention]
```

6.4 Doppler Lidar Backscatter: dl_XX_beta.txt

```
# ERF Doppler Lidar Simulator -- att. backscatter (1550 nm)
# Col 1: time [s],  Col 2: height [m],  Col 3: beta_att [m-1 sr-1]
```

6.5 MWR Profiles: `mwr_XX_profiles.txt`

```
# ERF MWR Simulator (Radiometrics MP-3000A)
# Col 1: time [s],  Col 2: height [m]
# Col 3: T [K],  Col 4: qv [g kg-1],  Col 5: qc [g kg-1]
```

6.6 MWR Column Integrals: `mwr_XX_column.txt`

```
# ERF MWR Simulator -- column integrals
# Col 1: time [s],  Col 2: LWP [g m-2],  Col 3: PWV [mm]
  1.000000e+01  2.534000e+01  1.872000e+01
```

6.7 MWR Brightness Temperatures: `mwr_XX_tb.txt` (optional)

```
# ERF MWR Simulator -- simulated brightness temperatures (35 channels)
# Col 1: time [s],  Col 2: frequency [GHz],  Col 3: TB [K]
  1.000000e+01  2.223500e+01  2.654300e+02
  1.000000e+01  2.250000e+01  2.671200e+02
  ...  (35 rows per timestep)
```

7 Software Architecture

7.1 Source File

File	Purpose
`Source/IO/ERF_InstrumentSimulators.H`	Header-only: all three simulator structs

7.2 Modified Files

File	Change
`Source/ERF.H`	Add `unique_ptr` members + method declarations
`Source/ERF.cpp`	Initialize in `InitData()`, call in `Evolve()` and `post_timestep()`
`Source/IO/Make.package`	Add `ERF_InstrumentSimulators.H` to `CEXE_headers`
`CMake/BuildERFExe.cmake`	Add `ERF_InstrumentSimulators.H` to target sources

7.3 Design Pattern

The simulator structs follow the `LineSampler` / `PlaneSampler` pattern in `Source/IO/ERF_SampleData.H`:

- `init()` reads `ParmParse` parameters and opens output files.
- `write()` checks interval, extracts z-columns from MultiFabs at the requested (i, j) location(s), computes derived quantities, and writes ASCII output.
- Column extraction is GPU-safe: `InstrumentSimUtil::extract_column()` and `extract_face_column()` use `amrex::ParallelFor` on the device followed by device-to-host copy and `MPI_Reduce` to the IO processor.

- All file I/O is guarded by `ParallelDescriptor::IOProcessor()`.
- The optional `InstrMicrophySizeData*` argument to the ceilometer and Doppler lidar `write()` functions carries pre-computed SBM or SDM extinction columns; `nullptr` invokes the bulk fixed-r_{eff} path.

7.4 Variable Access Summary

Quantity	ERF Access	Notes
ρ	`cons(i,j,k,Rho_comp)`	
$\rho\theta$	`cons(i,j,k,RhoTheta_comp)`	
ρq_v	`cons(i,j,k,RhoQ1_comp)`	
ρq_c	`cons(i,j,k,RhoQ2_comp)`	bulk liquid (all schemes)
ρq_i	`cons(i,j,k,RhoQ3_comp)`	bulk ice (guarded by `nComp()`)
u	`xvel(i,j,k)`	face-centred; averaged for cell-centre
v	`yvel(i,j,k)`	face-centred; averaged for cell-centre
w	`zvel(i,j,k)`	face-centred; averaged for cell-centre
$p_0(z)$	`base_state(i,j,k,BaseState::p0_comp)`	1D base pressure
SBM N_b	`cons[RhoQ1_comp + idx_nc_start + b]`	two-moment, [m^{-3}]
SBM r_b	`GetDiameterGridHostUM()[b] / 2`	precomputed table, [m]
SDM r_p	`soa.GetRealData(rtoff_r + radius)[p]`	per-particle radius [m]
SDM ξ_p	`soa.GetRealData(rtoff_r + multiplicity)[p]`	multiplicity [−]
SDM k_p	`p.idata(ERFParticlesIntIdxAoS::k)`	vertical cell index

8 Verification and Testing

1. **Build**: CMake build with `-DERF_ENABLE_MPI=ON` from an out-of-source build directory — should complete without new warnings.
2. **Functional**: Add an instrument block to any moist test input file (e.g., `Exec/RegTests/Bomex/inputs_Bomex`) run 100 steps, verify output files exist with monotonically increasing time values.
3. **Physics sanity**:
   - β_{att} peaks in the cloud layer and attenuates above it.
   - Ceilometer cloud base matches visual inspection of $q_c(z)$ profiles.
   - Doppler lidar w shows sign-changing turbulence in the boundary layer.
   - LWP from MWR $\approx \sum_k (\rho q_c)_k \Delta z_k$ (computed independently).
   - MWR $T(z)$ at the surface matches soundings and initial conditions.
4. **Microphysics sweep**: Run the same thermodynamic sounding with (a) a bulk scheme (e.g., Kessler or Morrison) and (b) SBM or SDM; compare β_{att} and LWP time series. Bulk results should be smooth functions of LWC; SBM/SDM results will differ in cloud-base onset timing and magnitude but should integrate to similar LWP.

References

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